Quantitative Data Analysis – Exercises

(Week 13)

In these exercises, you will learn:

- to perform simple and multiple linear regression analyses.
- to perform regression analysis based on regression trees and random forests.

In the data analysis process model, these exercises cover part of the steps "Statistical data analysis and/or Modeling" and "Evaluation & Interpretation" (see figure 1).

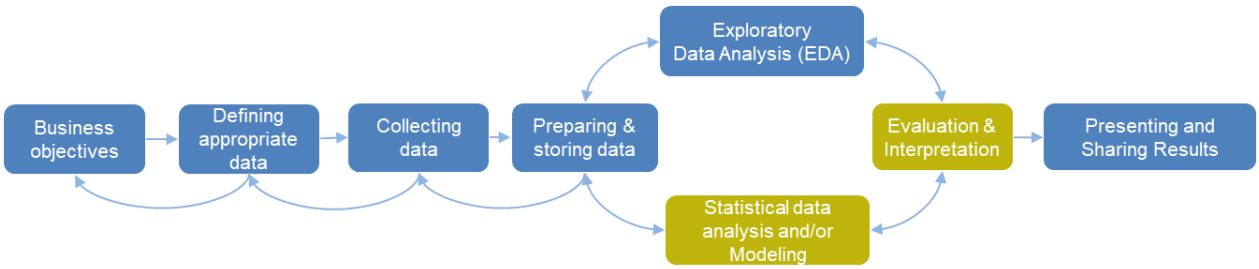


Figure 1: Data analysis process model (see slides of week 07)

Task 1

In this exercise, you will learn to perform simple and multiple linear regression analyses. The tasks are:

a) Run the Jupyter notebook 'linear_regression.ipynb' step by step and try to find out, what the Python code does.

b) Go to the section 'Simple linear regression …'. Create a new simple linear regression model with `price_per_m2` as target variable and `area` as the explanatory variable. Check the R-squared value of this new model. State whether the R-squared value is higher compared to the original model with `price` as target variable and `area` as the explanatory variable. Also check the histogram and the QQ-plot of the model residuals. State whether the residuals are approximately normally distributed or not.

c) Create a new simple linear regression model with price_per_m2 as target variable and rooms as the explanatory variable. Note: The new models explain a different target variable (price_per_m2 instead of price), i.e. the R-squared values describe different things and do not show which model predicts rental prices better. Check the R-squared value. State whether the R-squared value is higher compared to the original model with `price` as target variable and `area` as the explanatory variable? Also check the histogram and the QQ-plot of the model residuals. State whether the residuals are approximately normally distributed or not.

d) Go to the section 'Multiple linear regression …'. Include the variables `mean_taxable_income` and `dist_supermarket` as additional variables in the model. State whether the variables are statistically significant (at the 5% significance level). Note that this can be figured out by looking at the part of the output-table shown below. If the value **`P>|t| is smaller than 0.05, then`** a variable is statistically significant (at the 5% significance level).

	coef	std err	t	P>|t|	[0.025	0.975]
const	458.4991	65.979	6.949	0.000	328.937	588.061
area	15.0355	0.639	23.543	0.000	13.781	16.290
pop_dens	0.2381	0.012	20.189	0.000	0.215	0.261

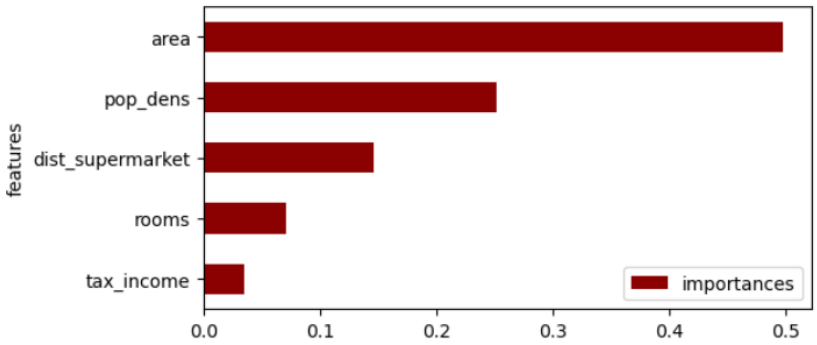
Expected results:

- The Jupyter notebook as html-file 'linear_regression.html' with the changes and short explanations according to b), c) and d).

Task 2

In this exercise, you will learn to perform regression analyses based on a regression tree and a random forest. The tasks are:

a) Run the Jupyter notebook 'regression_trees_random_forest.ipynb' step by step and try to find out, what the Python code does.
b) Go to the section 'Fit the regression tree model'. Change the model parameter `max_depth=3` to `max_depth=5`. This will change the depth (complexity) of the tree. Compare the output (text and graphic) with the output of the original regression tree (in which `max_depth=3`). Do you see any differences? In the Jupyter notebook, explain why.
c) Go to the section 'Calculate coefficient of determination …'. Look at the coefficient of determination (R-squared). Now, go to the section 'Create train and test samples for the regression tree …'. Set max_depth back to 3 and drop the two variables area and rooms from the train and test samples. Run the Jupyter notebook again. Does the R-squared value change? If so, explain why in the Jupyter notebook.
d) Go to the section 'Show feature importance' of the random forest and look at the barchart. Note that 'features' is another name for the 'explanatory variables' in a Machine Learning (ML) model. The barchart should look like this:



e) Go to the section 'Create train and test samples for the random forest …' and drop the variable area. Run the Jupyter notebook again.
f) Does the importance of features change? If yes, explain why in the Jupyter notebook. Note that feature importances are relative (they sum up to 1) and depend on the model and on the other variables in the model; they do not show causal effects.

Expected results:

- The Jupyter notebook as html-file 'regression_trees_random_forest.html' with the changes and short explanations according to b), c), d), e) and f).